CCCA Cell-State Analysis — Giustacchini et al. 2017, Hematologic

Chronic Myeloid Leukemia · SmartSeq2 (HiSeq 2000) · 2,040 malignant cells (35 patient plates)

GO / NO-GO GATES

Gate 1 — mixing & continuity — GO

Harmony(theta=6) integrates all 35 plates: cross-sample neighbour fraction 0.82; one connected component, 0 islands removed. Read on the regenerated embedding across an n_pcs/theta sweep, not a scalar: theta=6 folds the blast-crisis patient triad into the body (vs deleting ~180 cells at theta=2).

Gate 2 — module geography — GO

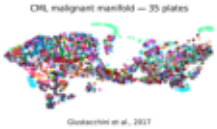
Non-branching case: a single (Gast/pSom)->Crest developmental gradient. Gast & pSom co-vary (smoothed r=+0.70), both oppose Crest (Gast<->Crest r=-0.93, pSom<->Crest r=-0.80). Along pseudotime Gast falls (rho -0.57), Crest rises to a terminal tip (rho +0.60, Crest-high decile pt=0.90). joint pSom&Crest enrichment 0.27 (<1 = inverse gradient, not a co-expression hub).

Gate 3 — cross-repression — NA

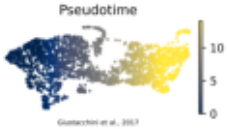
Not applicable — the manifold is non-branching (principal curve, 2 tips / 0 forks). No bifurcation, so the branching-only hub cross-repression test is not run (correlation / modules notebooks are not produced, per the deliverable spec).

STEP FIGURES

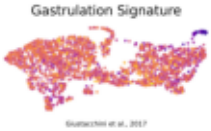
01 · Samples (Gate 1)



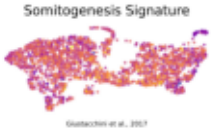
02 · Pseudotime



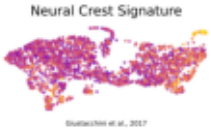
03 · Gastrulation



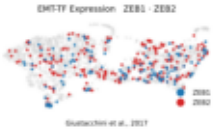
03 · Somitogenesis



03 · Neural Crest



04 · EMT-TFs



Dataset: CML (Giustacchini 2017), SmartSeq2 — 2,055 malignant BCR-ABL cells over 36 patient plates; single low-cycling body (~16% cycling), one continuous developmental axis.

Verdict: Pattern reproduces in the non-branching Linear-Gast form: one continuous gastrulation/somitogenesis -> neural-crest axis, Crest terminal. G1 GO / G2 GO / G3 NA. EMT-TFs ZEB1 & ZEB2 (only factors clearing detection), rising toward the Crest terminus.

PARAMETER CHOICES

Sample threshold	n>=20 cells (35/36 plates, 2,040 cells). >=100 floor rejected: SmartSeq2 plates cap at 154 cells, so a floor keeps only 3/36 plates (412 cells); no dominant multiplicative cliff -> low-end gap (15->22).
Overdispersed genes	5,909 (scFates find_overdispersed)
Integration	Harmony on sample, theta=6 (re-derived by stability sweep: folds blast-crisis patients into the body)
Embedding	n_pcs=20 (scree elbow: PC1 9%, PC2 1.8%, PC20-40 each <0.2%; n_pcs=40 fragments a patient speck)
Geometry	Non-branching -> scFates principal curve (ElPiGraph), Nodes=20, epg_lambda=0.3, epg_mu=100
Topology check	2 tips / 0 forks; fold self-approach 2.97 (>1 fold-free); orientation corr(x_lin, pseudotime)=+0.99
Root	curve tip 0 = Gast/pSom-high, Crest-low developmental origin (derived from signature order)
Module means (raw)	Gast +0.016 >= pSom +0.013 > Crest -0.016 (Linear-Gast); axis verified cell-cycle-free
EMT-TFs shown	ZEB1 (23% expr) & ZEB2 (42%); FOXC1/PRRX1/TWIST1/SNAI1 all <2% detected, gated out